# Project Design Phase-II
# Technology Stack (Architecture & Stack)

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| Date | 01 November 2026 |
| Team ID | [Enter Team ID] |
| Project Name | Global Olympic Sports Performance Analysis Using Tableau |
| Maximum Marks | 2 Marks |

## Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

| S.No | Component | Description | Technology | Justification / Purpose |
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| 1 | Programming / Query Language | Used for data cleaning, calculations, filtering and analysis. | Tableau Calculated Fields; Python (optional); SQL (optional) | Tableau calculations create medal counts, distinct medal keys and efficiency measures. Python/SQL may support preprocessing if needed. |
| 2 | Database / Data Warehouse | Stores source and supporting lookup data. | CSV files / Excel; Tableau data source | The primary athlete_events.csv file is a structured CSV. Additional CSV/Excel tables are needed for region, host country and sports investment. |
| 3 | Data Ingestion | Loads source data into the analytical environment. | Tableau Text File connection; Excel connection | Connects CSV and lookup files directly to Tableau for analysis. |
| 4 | Data Transformation and Business Logic | Cleans, relates, filters and calculates analytical measures. | Tableau Data Source page; Tableau calculated fields; optional Tableau Prep | Supports data type checks, null handling, relationships, year/season filters, distinct medal counting and medal-efficiency calculations. |

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| 5 | Data Visualization / Analytics | Creates interactive charts, dashboards and stories. | Tableau Desktop / Tableau Public | Builds nine visualizations across three dashboards and a four-point story for the three stakeholder scenarios. |
| 6 | Front-End / User Interface | Provides access to interactive insights. | Tableau Public/Server embedding; HTML, CSS and Flask (planned) | Users can view dashboards through a simple Flask web interface, subject to publishing and embedding permissions. |